FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Physics — Class 9 (Science Group)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till Ex. 6.4 only) | Unit 9 — Nature of Science & Physics

Total Marks: 43 **Time: 2 Hours** **Paper: Exam-14 (Version 1)**

SECTION A — MCQ Answer Key (24 × 1 = 24 Marks)

#	Question	Correct Answer	Explanation	Source in her textbook
1	Least count of a screw gauge = pitch divided by:	**C. number of circular scale divisions** 1	The book states the least count of a screw gauge is found by dividing its pitch by the number of circular scale divisions. With pitch 0.5 mm and 50 divisions, L.C. = 0.01 mm.	*U1 §1.7.3; SLO P-09-A-19*
2	0.00085 m in scientific notation	**A. 8.5×10^{-4} m** 1	Move the decimal point 4 places to the right to get 8.5, so the power of ten is −4. Standard form needs one non-zero digit before the decimal, which rules out C and D.	*U1 §1.4; SLO P-09-A-05*
3	Which is a vector quantity?	**D. momentum** 1	Momentum is listed in the SLO among the vector quantities. Mass, temperature and energy are all scalars, having magnitude only.	*U1; SLOs P-09-A-08 / A-09*
4	SI prefix for 10^{-9}	**C. nano** 1	From the prefix table: nano = 10^{-9}, micro = 10^{-6}, milli = 10^{-3}, mega = 10^{6}.	*U1 §1.5, Table 1.3; SLO P-11-A-06*
5	Area under a speed-time graph	**B. distance travelled** 1	Speed × time = distance, so the area beneath the line gives the distance covered. The gradient, not the area, gives acceleration.	*U2; SLOs P-09-B-12 / B-13*
6	Free-fall acceleration near Earth	**A. 9.8 m/s^2** 1	The approximate value of the acceleration of free fall near the Earth's surface is 9.8 m/s^2, directed downwards.	*U2; SLO P-09-B-10*
7	Velocity changes by unequal amounts in equal times	**D. non-uniform** 1	Uniform acceleration means equal changes of velocity in equal intervals of time. Unequal changes therefore mean the acceleration is non-uniform.	*U2; SLO P-09-B-08*
8	96 m in 6 s, average speed	**B. 16 m/s** 1	Average speed = total distance / total time = 96	*U2; SLO P-09-B-04*

			m ÷ 6 s = 16 m/s.	
9	Newton's second law: acceleration is	**C. directly proportional to the applied force** 1	From F = ma, a = F/m. Acceleration is directly proportional to the applied force and inversely proportional to the mass, which rules out A and B.	*U3 §3.2.2; SLO P-09-B-30*
10	SI unit of impulse	**D. newton second (N s)** 1	Impulse = force × time, so its unit is N × s = N s. It equals the change in momentum, which is also measured in N s (or kg m/s).	*U3 §3.6; SLO P-09-B-40*
11	Newton's laws cannot readily be applied on the	**A. atomic scale** 1	The book states Newton's laws are very useful for everyday objects but are not readily applied on the very small scale, where position and acceleration are not well defined and quantum mechanics takes over.	*U3 §3.2.4; SLO P-09-B-33*
12	Instrument used to measure mass	**B. electronic balance** 1	An electronic balance measures mass. A force meter (spring balance) measures weight, a measuring cylinder measures volume and a stopwatch measures time.	*U3; SLOs P-09-B-21 / B-22*
13	Point at which the whole weight appears to act	**C. centre of gravity** 1	The centre of gravity is the single point at which the entire weight of a body appears to act. The centre of mass is where the whole mass appears to be concentrated.	*U4 §4.3; SLO P-09-B-47*
14	Average orbital speed of a satellite	**A. v = 2πr / T** 1	In one complete orbit the satellite travels the circumference 2πr in a time equal to the orbital period T, so average orbital speed = 2πr / T.	*U4 §4.8.1; SLO P-09-F-01*
15	Two parallel forces in the same direction	**D. like parallel forces** 1	Parallel forces acting in the same direction are called like parallel forces. If they act in opposite directions they are unlike parallel forces.	*U4 §4.2; SLO P-09-B-43*
16	Principle of moments condition	**B. equal to the sum of the anticlockwise moments** 1	The principle of moments (the second condition of equilibrium) states that for a body in equilibrium the sum of clockwise moments about a point equals the sum of anticlockwise moments about the same point.	*U4 §4.4.2; SLO P-09-B-45*
17	SI unit of spring constant	**A. newton per metre (N/m)** 1	From Hooke's law F = kx, k = F/x, so the unit is	*U5 §5.1; SLO P-09-B-56*

			newton divided by metre, N/m. The worked example in the book gives k = 4,900 N/m.	
18	Point where extension stops being proportional to load	**C. limit of proportionality** 1	Up to the limit of proportionality the load-extension graph is a straight line through the origin and extension is directly proportional to load. Beyond it the line curves.	*U5; SLO P-09-B-58*
19	Instrument for atmospheric pressure	**B. barometer** 1	A liquid (mercury) barometer measures atmospheric pressure. A manometer measures the pressure of an enclosed fluid or gas supply.	*U5 §5.3.1; SLO P-09-B-83*
20	Mercury column supported at sea level	**D. 76 cm** 1	At sea level the atmosphere supports a mercury column of about 760 mm, which is 76 cm. The book gives 1 atm = 760 mmHg = 101.325 kPa.	*U5 §5.3.1; SLO P-09-B-83*
21	Which is non-renewable?	**B. coal** 1	Coal is a fossil fuel, limited in supply and not replaced naturally in a short time. Wind, tidal and geothermal are all renewable resources.	*U6 §6.3; SLO P-09-B-70*
22	One watt equals	**C. 1 joule per second** 1	Power is the rate of doing work, P = W/t, so 1 watt = 1 joule / 1 second, written 1 W = 1 J s^{-1}.	*U6 §6.4; SLO P-09-B-73*
23	Why a perpetual motion machine is impossible	**A. conservation of energy** 1	A perpetual motion machine is a hypothetical machine that would do work indefinitely with no external energy source. That would create energy from nothing, which the principle of conservation of energy forbids.	*U6 §6.2; SLOs P-09-B-68 / B-69*
24	A claim that can be shown to be wrong by evidence	**D. falsifiability** 1	Falsifiability is the idea that a scientific statement must be testable in a way that could in principle prove it false. A claim that no evidence could disprove is not scientific.	*U9 §9.3; SLO P-10-G-07*

SECTION B — Short Questions (3 × 3 = 9 Marks)

Q.2 (i) Vernier caliper: least count and reading 3 marks

(a) Least count 1

Least Count = (smallest division on main scale) ÷ (total number of divisions on vernier scale)

L.C. = 1 mm ÷ 50 = **0.02 mm = 0.002 cm**

(b) Length of the rod 2

Length = main scale reading + (coinciding vernier division × least count) 1

Length = 6.7 cm + (25 × 0.002 cm) = 6.7 cm + 0.05 cm

Length = 6.75 cm 1

Source: U1 §1.7.2 (least count formula and the method for taking a vernier reading); SLOs P-09-A-12, P-09-A-19. Accept 0.02 mm or 0.002 cm for (a). Award the working mark in (b) even if the arithmetic slips.

Q.2 (ii) Centre of mass and centre of gravity 3 marks

(a) Difference 2

Centre of mass	Centre of gravity
The point at which the whole mass of the body may be considered to be concentrated. 1	The single point at which the whole weight of the body appears to act. 1

(b) When they are not the same 1

For small objects the centre of mass and the centre of gravity are at the same point. They are **not** the same for very large objects, because the value of g (the gravitational field strength) is not the same at every part of such a body. 1

Source: U4 §4.3, including the sentence "The center of mass and center of gravity (CM/CG) are same for small objects"; SLO P-09-B-47. Accept any correct wording of "very large / very tall body where g varies".

Q.2 (iii) Barometer with a different liquid 3 marks

(a) Height of the liquid column 2

The atmospheric pressure is unchanged, and for a liquid column P = ρgh, so ρh must be the same for both liquids:

$\rho_1 h_1 = \rho_2 h_2$ 1

$h_2 = (\rho_1 h_1) \div \rho_2 = (13{,}600\ \text{kg/m}^3 \times 0.76\ \text{m}) \div 3{,}400\ \text{kg/m}^3$

$h_2 = 10{,}336 \div 3{,}400 =$ **3.04 m** 1

(b) Dependence on density 1

The height of the column is **inversely proportional to the density** of the liquid: the lower the density, the taller the column needed to balance the same atmospheric pressure. (Here the density is one quarter that of mercury, so the column is four times as tall.) 1

Source: U5 §5.3.1 (mercury barometer, 76 cm at sea level) and §5.4 (P = ρgh); SLOs P-09-B-83, P-09-B-84. The book's own review question asks the same idea in reverse ("if a liquid has density twice that of mercury, what will be the height of the liquid column").

SECTION C — Long Questions (2 × 5 = 10 Marks)

Q.3 (i) Speed-time graph **5 marks**

(a) Description of the motion **1**

Stage A (0 s to 10 s): the body starts from rest and accelerates uniformly to 14 m/s.

Stage B (10 s to 30 s): the body moves with constant (uniform) speed of 14 m/s, so its acceleration is zero.

Stage C (30 s to 40 s): the body decelerates uniformly from 14 m/s and comes to rest. **1**

(b) Acceleration during stage A **2**

a = (v − u) ÷ t **1**

a = (14 m/s − 0 m/s) ÷ 10 s = 14 ÷ 10

a = 1.4 m/s^2 **1**

(c) Total distance from the area under the graph **2**

Stage A (triangle) = ½ × base × height = ½ × 10 s × 14 m/s = 70 m

Stage B (rectangle) = 20 s × 14 m/s = 280 m **1**

Stage C (triangle) = ½ × 10 s × 14 m/s = 70 m

Total distance = 70 + 280 + 70 = **420 m** **1**

Source: U2 §2.6 (speed-time graphs, gradient gives acceleration, area gives distance); SLOs P-09-B-09, P-09-B-12, P-09-B-13, P-09-B-14. Accept the trapezium method ½(sum of parallel sides) × height = ½(40 + 20) × 14 = 420 m for full marks in (c).

Q.3 (ii) Energy resources and power **5 marks**

(a) Renewable and non-renewable energy resources **2**

A **non-renewable** energy resource is one that is limited and cannot be replaced naturally in a short time; once used it is gone. Examples: coal, natural gas, petroleum, nuclear fuel (uranium). **1**

A **renewable** energy resource is one that is replaced by an ongoing natural process, so it cannot simply run out. Examples: solar (sunshine), wind, flowing water (hydroelectric), geothermal, biomass, tidal, wave. **1**

(Any two correct examples of each are enough.)

(b) Hydroelectric power **1**

Advantage: it is renewable and efficient, and once the facilities are built and running they have minimal impact on the environment (no fuel cost, no smoke or greenhouse gases).

Disadvantage: it requires fast-flowing water, and large facilities often cause ecological damage (flooding land behind the dam, high initial cost). **1**

(c) Work and power of the conveyor belt **2**

The crate moves at a steady speed, so the belt's driving force balances the friction, F = 350 N.

Distance moved: d = v × t = 2 m/s × 90 s = 180 m

Work done: W = F × d = 350 N × 180 m = **63,000 J = 6.3 × 10^4 J** **1**

Power: P = W ÷ t = 63,000 J ÷ 90 s = **700 W** **1**

(P = F × v = 350 × 2 = 700 W gives the same result and earns full marks.)

Source: U6 §6.3 (definitions of renewable and non-renewable resources), §6.3.4 (hydroelectric: "renewable and efficient... however, generating power from water requires fast-flowing water, and it often causes ecological damage"), §6.1 (W = F × d) and §6.4 (P = W/t); SLOs P-09-B-61, P-09-B-70, P-09-B-72, P-09-B-73. Note: efficiency (§6.5) is outside the studied syllabus and is not examined anywhere on this paper.

—— End of Answer Key ——